$$\xrightarrow{R_1 \to R_1 - \frac{1}{2}R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 2 & -1 & 0 \\ 0 & 1 & -1 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow[R_2 \to R_2 + R_3]{R_1 \to R_1 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -1 & -1 \\ 0 & 1 & 0 & -4 & 2 & 1 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right].$$

So, $A^{-1} = \begin{bmatrix} 3 & -1 & -1 \\ -4 & 2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$.

EXERCISE 1.2

1. Find the rank of the following matrices by minor method:

(i) $\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & -6 \\ 5 & 1 & -1 \end{bmatrix}$ (v) $\begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ 8 & 1 & 0 & 2 \end{bmatrix}$

2. Find the rank of the following matrices by row reduction method:

(i) $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}$ (iii) $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$

3. Find the inverse of each of the following by Gauss-Jordan method:

(i) $\begin{bmatrix} 2 & -1 \\ 5 & -2 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 6 & -2 & -3 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

1.4 Applications of Matrices: Solving System of Linear Equations

One of the important applications of matrices and determinants is solving of system of linear equations. Systems of linear equations arise as mathematical models of several phenomena occurring in biology, chemistry, commerce, economics, physics and engineering. For instance, analysis of circuit theory, analysis of input-output models, and analysis of chemical reactions require solutions of systems of linear equations.

1.4.1 Formation of a System of Linear Equations

The meaning of a system of linear equations can be understood by formulating a mathematical model of a simple practical problem.

Three persons A, B and C go to a supermarket to purchase same brands of rice and sugar. Person A buys 5 Kilograms of rice and 3 Kilograms of sugar and pays ₹ 440. Person B purchases 6 Kilograms of rice and 2 Kilograms of sugar and pays ₹ 400. Person C purchases 8 Kilograms of rice and 5 Kilograms of sugar and pays ₹ 720. Let us formulate a mathematical model to compute the price per Kilogram of rice and the price per Kilogram of sugar. Let x be the price in rupees per Kilogram of rice and y be the price in rupees per Kilogram of sugar. Person A buys 5 Kilograms of rice and 3 Kilograms sugar and pays ₹ 440. So, $5x + 3y = 440$. Similarly, by considering Person B and Person C, we get $6x + 2y = 400$ and $8x + 5y = 720$. Hence the mathematical model is to obtain x and y such that

$$5x + 3y = 440, \quad 6x + 2y = 400, \quad 8x + 5y = 720.$$

12th_Maths_EM_Vol1_CH 1_Matrices.indd 27 16-12-2022 12:17:55